
INDEINITE INTERATION
Indefinit
Integration

INDEFINIT INTEGRATION

if $f(x)$ is a continuous as well as differentiable f^n, then $\int f(x) \cdot dx = F(x) + c$.

$\Rightarrow$ the antiderivative (integration) of $f(x)$ will be $F(x) + c$.

$$\int f(x) \cdot dx = F(x) + c$$

- $f(x)$: integrand
- dx : variable of integration
- $F(x)$: integral (primitive)
- c : integration constant

WORKING RULES

1. if $\int f(x) \cdot dx = F(x) + c$, then

$$\int a f(bx \pm k) \cdot dx = \frac{a}{b} F(bx \pm k) + c$$

2. $\int f(x) \pm g(x) \cdot dx = \int F(x) \cdot dx \pm \int g(x) \cdot dx$

3. $\frac{d}{dx}\left(\int f(x) \cdot dx\right) = f(x)$

4. $\int \frac{d}{dx}\left(f(x)\right) \cdot dx = f(x) + c$

STANDARD FORMULAS

- $\int k \cdot dx = kx + c$
- $\int x^n dx = \frac{x^{n+1}}{n+1} + c$
- $\int e^x dx = e^x + c$
- $\int a^x dx = \frac{a^x}{\ln a} + c$
- $\int \ln x \cdot dx = x(\ln x - 1) + c$
- $\int \frac{1}{x} \cdot dx = \ln x + c$
- $\int \sin x \, dx = -\cos x + c$
- $\int \cos x \, dx = \sin x + c$
- $\int \tan x \, dx = -\ln|\cos x| + c$, $\ln|\sec x| + c$
- $\int \cot x \, dx = \ln|\sin x| + c$, $-\ln|\operatorname{cosec} x| + c$
- $\int \sec x \, dx = \ln|\sec x + \tan x| + c$
- $\int \operatorname{cosec} x \cdot dx = \ln|\operatorname{cosec} x - \cot x| + c$
- $\int \sec^2 x \, dx = \tan x + c$
- $\int \operatorname{cosec}^2 x \, dx = -\cot x + c$
- $\int \sec x \tan x \, dx = \sec x + c$
- $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + c$
- $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln\left|\frac{x-a}{x+a}\right| + c$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln\left|\frac{x+a}{a-x}\right| + c$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$$

$$\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left(x + \sqrt{x^2-a^2}\right) + c$$

$$\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left(x + \sqrt{x^2+a^2}\right) + c$$

$$\int \sqrt{x^2+a^2}\cdot dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right) + c$$

$$\int \sqrt{x^2-a^2}\cdot dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left(x+\sqrt{x^2-a^2}\right) + c$$

$$\int \sqrt{a^2-x^2}\cdot dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + c$$

Question $\int 2^x e^x dx$

$$\int (2e)^x dx = \frac{2e^x}{\ln(2e)} + c$$

Question $\int \frac{1-\tan^2 x}{1+\tan^2 x}\cdot dx$

$$\int \sin 2x \cdot dx = \frac{\sin 2x}{2} + c$$

Question $\int \frac{1+\cos^2 x}{1+\cos 2x} \cdot dx$

$$= \int \frac{1+\cos^2 x}{2\cos^2 x} \cdot dx = \frac{1}{2}\int (\sec^2 x + 1)\, dx$$

$$= \frac{1}{2}(\tan x + x) + c$$

Question $\int \frac{\cos 2x}{\cos^2 x \sin^2 x}$

$$= \frac{4\cos 2x}{(\sin 2x)^2} = \int 4\cot 2x \cdot \operatorname{cosec} 2x$$

$$= -\frac{4\operatorname{cosec} 2x}{2} + c$$

$$= -2\operatorname{cosec} 2x + c$$

$$\int \frac{\cos^2 x - \sin^2 x}{\cos^2 x \sin^2 x} \cdot dx = \int \operatorname{cosec}^2 x\, dx - \int \sec^2 x\, dx$$

$$= -\cot x - \tan x + c$$

Question $\int \frac{1+2x^2}{x^2(1+x^2)} \cdot dx$

$$\frac{1+x^2}{x^2(1+x^2)} + \frac{x^2}{x^2(1+x^2)}$$

$$\int \frac{1}{1+x} = \tan^{-1} x$$

$$\int x^{-2} + (1+x^2)^{-1} \cdot dx$$

$$= \frac{x^{-1}}{-1} + \tan^{-1} x + c$$

Question $\int \sin x \cdot \sin 7x \, dx$

$$\frac{-2 \sin x \sin 7x}{-2} = \int \frac{\cos(6x) - \cos 8x}{-2}$$

$$= \frac{\sin 6x}{2 \times 6} - \frac{\sin 8x}{2 \times 8} + c$$

Question $\int \frac{(1+x)^2}{x^2(1+x^2)} \cdot dx$

$$= \int \frac{1+x^2+2x}{x^2(1+x^2)} dx = \int \frac{dx}{x^2} + \int \frac{2}{x(1+x^2)} \cdot dx$$

$$= \frac{-1}{x} + \int \frac{2x\,dx}{x^2(1+x^2)} \cdot dx$$

$\{1+x^2 = t\}$
$\{2x\,dx = dt\}$

$$\Rightarrow \frac{-1}{x} + \int \frac{dt}{(t-1)t}$$

$$\Rightarrow \frac{-1}{x} - \int \frac{t-1-t}{t(t-1)} \cdot dt$$

$$\Rightarrow \frac{-1}{x} - \int \frac{dt}{t} + \int \frac{dt}{t-1}$$

$$\Rightarrow \frac{-1}{x} - \ln|t| + \ln|t-1| + c$$

$$\Rightarrow \frac{-1}{x} - \ln|x^2+1| + \ln|x^2| + c$$

Question

$$\int \frac{2x-1}{x-2}\,dx$$

$$\int \frac{x-1+x-1+1}{x-2} = \frac{2x-4+3}{x-2}$$

$$= \int 2\,dx + \int \frac{3}{x-2}\,dx$$

$$= 2x + 3\ln|x-2| + c$$

Question

$$\int \frac{x^6-1}{x^2+1}\,dx$$

(I) $$\frac{(x^2+1)(x^4+1+x^2)}{x^2+1} = \frac{(x^2-1)\,x^2\,(x^2+1)}{x^2+1} + \frac{1}{x^2+1}$$

$$= \int x^4\,dx - \int x^2\,dx + \int \frac{1}{x^2+1}\,dx$$

II $$\frac{(x^2)^3+1-2}{x^2+1}\,dx = \frac{x^5}{5} - \frac{x^3}{3} - 2\tan^{-1}x + c$$

Question

$$\int \frac{(x^2+\sin^2x)\sec^2x}{1+x^2}\,dx$$

$$\int \frac{(x^2+\sin^2x+1-1)\sec^2x}{1+x^2}\,dx = \int \frac{\sec^2x(1+x^2)}{1+x^2}\,dx$$

$$+ (-)\left[\frac{\cos^2x\sec^2x}{1+x^2}\,dx\right]$$

$$= \tan x - \tan^{-1}x + c$$

Question

$$\int \frac{\cos x}{\cos(x-a)} \cdot dx$$

$$\frac{\cos(x+a-a)}{\cos(x-a)} \cdot dx$$

$$\frac{\cos x}{\cos a} \int \frac{\cos(x+a)\cos a + \sin(x+a)\sin a}{\cos(x-a)}$$

$$= \int \cos a \cdot dx - \int \sin a \cdot \tan(x-a) \cdot dx$$

$$= x\cos a - \sin a \ln|\sec(x-a)|$$

Question

$$\int \frac{dx}{1+\sin x} \qquad \int \frac{1-\sin x}{\cos^2 x} \cdot dx$$

$$\int (\sec^2 x - \tan x \sec x) \cdot dx$$

$$= \tan x - \sec x + c$$

Question

$$\int \frac{2+3x^2}{x^2(1+x^2)} \cdot dx$$

$$= \int \frac{2+2x^2}{x^2(1+x^2)} + \frac{x^2}{x^2(1+x^2)} \cdot dx$$

$$= \left(\frac{2}{x^2} + 6 + \frac{1}{1+x^2}\right) \cdot dx$$

$$= \frac{-2}{x} + 6 + \tan^{-1} x + c$$

Question $\int \frac{\cos 8x - \cos 7x}{1 + 2\cos 5x} \cdot dx$

Question $\int \frac{x^2 + x^2 + 1}{2(1+x^2)} \cdot dx$

$$\frac{x^4 + 2x^2 + 1}{2(1+x^2)} + \frac{x^2}{2(1+x^2)}$$

$$\int \frac{(1+x^2)}{2} - \frac{1}{2}\frac{(x^2+1)}{1+x^2} - \frac{1}{2(1+x^2)} \cdot dx$$

$$= \frac{x}{2} + \frac{x^3}{6} - \frac{x}{2} + \frac{\tan^{-1}x}{2} + c$$

$$= \frac{x^3}{6} + \frac{\tan^{-1}x}{2} + c$$

METHOD OF INTEGRATION

SUBSTITUTION METHOD

DIRECT SUBSTITUTION

if Question is in the form of

$$\int f(g(x)) \cdot g'(x) \cdot dx$$

assume $g(x) = t$

$g'(x) \cdot dx = dt$

$\int f(t)\,dt = F(t) + c$

$= F(g(x)) + c$

Question $\int \sin x^2 \cdot x\,dx$

$x^2 = t \qquad 2x\,dx = dt \qquad x\,dx = \frac{dt}{2}$

$\frac{1}{2}\int \sin t\,dt = \frac{1}{2}(-\cos t) + c$

$= -\frac{\cos x^2}{2} + c$

Question $\int \frac{\cos(\ln x)}{x} \cdot dx$

$= \sin(\ln x) + c$

Question $\int \frac{\cos(\sin^{-1} x)}{\sqrt{1-x^2}} \cdot dx$

$= \sin(\sin^{-1} x) + c = x + c$

Question $\int \frac{dx}{x+\sqrt{x}} = \int \frac{x-\sqrt{x}}{x^2-x}\,dx = \int \frac{x-\sqrt{x}}{x(x-1)} \cdot dx$

$\int \frac{dx}{\sqrt{x}(1+\sqrt{x})} \qquad \sqrt{x} = t$

$\frac{1}{2\sqrt{x}} \cdot dx = dt$

$$\int \frac{2\,dt}{(1+t)} = 2\cdot \ln(1+t) + c$$

$$= 2\ln(1+\sqrt{x}) + c$$

Question $\int \cos^3 x \sin^2 x\, dx$

$\cos x = t$

$-\sin x\, dx = dt$

$\cos^2 x \cdot \sin x$

$= -2\cos x \sin x$

$\frac{4(\cos x \sin x)^2 \cdot \cos x}{4}$

$\frac{(\sin 2x)^2 \cos x}{4}$

$\frac{\cos^2 x\,(2\cos x \sin x)}{2} - \sin x$

$\cos^2 x = t$

$2\cos x \sin x\, dx = dt$

$\frac{-t)\,dt\,\sqrt{1-t^2}}{2}$

$\sin x = t$

$\cos x \cdot dx \cdot dt$

$\frac{1}{2}\int \sqrt{1-t^2}\cdot t$

$\int (1-t^2)\,dt$

$= \int t^2\,dt - \int t^4\,dt$

$= \frac{t^3}{3} - \frac{t^4}{4} + c$

Question $\int \frac{\tan\sqrt{x}\,\sec^2\sqrt{x}}{\sqrt{x}}\cdot dx$

$\tan\sqrt{x} = t$

$\sec^2\sqrt{x}\cdot\frac{1}{x\sqrt{x}}\cdot dx = 2\,dt$

$$2\int t\,dt = 2\frac{t^2}{2} + c = \frac{2(\tan\sqrt{x})^2}{2} + c$$

Question $\int \frac{\tan(\tan x - x)\sin^2 x}{1-\sin^2 x}\,dx$

5. $\int \frac{\sqrt{\tan x}}{\sin 2x}\,dx$

Question $\int \frac{dx}{e^x+1}$

Question $\frac{d}{dx}(\tan x - x) = \sec^2 x - 1 = \tan^2 x$

$$\int \frac{\tan(\tan x - x)\tan^2 x\,dx}{g(f(x))}\cdot f'(x)$$

$$= h(f(x)) + c$$

$$= \ln|\sec(\tan x - x)| + c$$

Question $\int \frac{1}{e^x+1}\,dx = \int \frac{e^x+1}{e^x+1}\,dx - \int \frac{e^x}{e^x+1}\,dx$

$$= x - \ln(e^x+1) + c$$

If a f^n is given in such a way that

$$\int \frac{a\sin x + b\cos x}{c\sin x + d\cos x} \cdot dx$$

$$a\sin x + b\cos x = A(c\sin x + d\cos x) + B(c\cos x - d\sin x)$$

to obtain the value of $A + B$.

Compare the ~~diff~~ coeff. of $\sin x$ & $\cos x$.

$$= \int \frac{A(c\sin x + d\cos x) + B(c\cos x - d\sin x)}{c\sin x + d\cos x} \cdot dx$$

$$\Rightarrow A\int dx + B\int \frac{c\cos x - d\sin x}{c\sin x + d\sin x} \cdot dx$$

$$\Rightarrow Ax + B\ln|c\sin x + d\sin x| + C$$



$$\int \frac{\sin x + 3\cos x}{5\sin x - \cos x} \cdot dx$$

$$\sin x + 3\cos x = A(5\sin x - \cos x) + B(5\cos x + \sin x)$$

$$1 = 5A + B \qquad = 3(5B - A)5$$

$$26B = 16 \Rightarrow B = \frac{8}{13}$$

$$A = \frac{1}{13}$$

$$\frac{2}{13}x + \frac{8}{13}\ln|5\sin x - 8\cos x| + C$$

Question

$$\int \frac{dx}{1+\tan x} = \int \frac{2\sin x + \cos x - \sin x}{\cos x + \sin x}$$

$$\int \frac{\cos x + \sin x - \sin x}{\sin x + \cos x} = \int \frac{\cos x + 0\sin x}{\cos x + \sin x} \cdot dx$$

$$\cos x = A(\sin x + \cos x) + B(\cos x - \sin x)$$

$$0 = A - B \qquad A = B \qquad A + B = 1$$

$$2B = 1$$

$$B = \frac{1}{2}$$

$$\Rightarrow \frac{1}{2}x + \frac{1}{2}\ln|\sin x + \cos x| + c$$

Question

$$\int \frac{\sin x}{\sin x + \cos x} \cdot dx$$

$$\frac{x}{2} - \frac{1}{2}\ln|\sin x + \cos x| + c$$

$$\int \frac{\sqrt{x}\,.dx}{1+x^3}$$

$$x^{3/2} = t \Rightarrow \frac{3}{2}x^{1/2}\,dx = dt$$

$$\sqrt{x}\,dt = \frac{2}{3}\,dt$$

$$\int \frac{2}{3}\frac{dt}{(1+t^2)} = \frac{2}{3}\tan^{-1}(t) + c$$

INDIRECT SUBSTITUTION

$\int f(x).dx \Rightarrow f(x)$ as a product of 2 function $f_1(x)$ and $f_2(x)$ in such a way that one f^n will become the differentiable of function f^n i.e.



$$\int \frac{\cos x - \sin x}{3 + 4\sin 2x}.dx$$

$$\sin x + \cos x = t$$

$$\sin 2x = (\sin x + \cos x)^2 - 1$$

$$\int \frac{\cos x - \sin x}{3 + 4(\sin x + \cos x)^2 - c}.dx$$

$$= \frac{1}{4}\int \frac{dt}{t^2 - \left(\frac{1}{2}\right)^2}$$

$$= \frac{1}{4}.\frac{1}{2\left(\frac{1}{2}\right)} \ln\left|\frac{t - 1/2}{t + 1/2}\right| + c$$



$$\int \frac{\sin x + \cos x}{4 - 5\sin 2x}.dx$$

Let $\sin x - \cos x = t$

$$\Rightarrow \int \frac{dt}{-1 + 5t^2}$$

$$\Rightarrow -\frac{1}{5}\int \frac{dt}{t^2 - \left(\frac{1}{\sqrt{5}}\right)^2}$$

$$\Rightarrow \frac{1}{5}.2\left(\frac{1}{\sqrt{5}}\right) \tan^{-1}\left(\frac{t - 1/\sqrt{5}}{t + 1/\sqrt{5}}\right)$$

$4 - 5(1 - t^2)$

$4 - 5 + 5t^2$

$1 + 5t^2$

$$\Rightarrow \frac{\sqrt{5}}{10\sqrt{5}} \tan^{-1}\left(\frac{(\sin x-\cos x)^{1/\sqrt{5}}}{\sin x-\cos x+\frac{1}{\sqrt{5}}}\right)+c$$

$$\sin^2 x+\cos^2 x+\sin 2x$$

$$=(\sin x+\cos x)^2$$

$$=1+\sin 2x$$

$$(\sin x-\cos x)^2=1-\sin 2x \text{ can solve further.}$$

Question

$$\int \sqrt{\frac{x^3}{x^5+1}}\cdot dx$$

$$=\int \frac{x^{3/2}}{\sqrt{x^5+1}}\cdot dx = \int \frac{2}{5}\frac{dt}{\sqrt{t^2+1}}$$

$$=\frac{2}{5}\ln\left|t^2+\sqrt{t^2+1}\right|$$

$$=\frac{2}{5}\ln\left|x^{5/2}+\sqrt{x^5+1}\right|+c$$

Question

$$\int \frac{\sin x\,dx}{1+2\sin 2x}$$

{ Twin Integration } ...

$$\rightarrow \frac{1}{2}\int \frac{2\sin x\,dx}{1+2\sin 2x} = \frac{1}{2}\int \frac{(\sin x+\cos x)(\cos x-\sin x)}{1+2\sin 2x}$$

$$\Rightarrow \frac{1}{2}\int \frac{\sin x+\cos x}{1+2\sin 2x}\cdot dx - \frac{1}{2}\int \frac{\cos x-\sin x}{1+2\sin 2x}\cdot dx$$

$\sin x-\cos x=t$

$2\sin 2x=2\left[1-(\sin x-\cos x)\right]$

$2\sin 2x=2(1-t^2)$

$\sin x+\cos x=t$

$2\sin 2x=2(t^2-1)$

$$\Rightarrow \frac{1}{2}\int \frac{1}{3-2t^2}\cdot dt \qquad \Rightarrow -\frac{1}{2}\int \frac{dt}{2t^2-1}$$

$$\Rightarrow -\frac{1}{2}\int \frac{1}{2\left(t^2-3/2\right)}\cdot dt \qquad \Rightarrow -\frac{1}{4}\int \frac{dt}{t^2-\left(\frac{1}{\sqrt{2}}\right)^2}$$

$$\Rightarrow -\frac{1}{4}\int \frac{1}{t^2-\left(\sqrt{3}/2\right)^2}\cdot dt \qquad \ln\left[\frac{t-1/\sqrt{2}}{t+1/\sqrt{2}}\right]$$

$$\Rightarrow -\frac{1}{4}\left[\frac{1}{2}\frac{\sqrt{2}}{\sqrt{3}}\right]\left[\ln\frac{t-\sqrt{3}/2}{t+\sqrt{3}/2}\right] \qquad \Rightarrow \frac{-1}{4}\cdot\frac{1}{2\left(\frac{1}{\sqrt{2}}\right)}$$

Question

$$\int \frac{x^2-1}{x^4+5x^2+1}\cdot dx$$

$$\int \frac{1-\frac{1}{x^2}}{x^2+5+\frac{1}{x^2}}\cdot dx = \int \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)^2+\left(\sqrt{3}\right)^2}\cdot dx$$

$$x+\frac{1}{x}=t$$

$$\left(1-\frac{1}{x^2}\right)\cdot dx = dt$$

$$\Rightarrow \int \frac{dt}{t^2+(\sqrt{3})^2} = \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{t}{\sqrt{3}}\right)+c$$

Question

$$\int \frac{(\sin x+1)\cos x}{\sin^4 x+5\sin^2 x+1}\cdot dx$$

$$\sin x = t$$

$$\Rightarrow \int \frac{(t^2+1)\cdot dt}{t^4+5t^2+1} \Rightarrow \int \frac{\left(1+1/t^2\right)\cdot dt}{\left(t-\frac{1}{t}\right)^2+(\sqrt{7})^2}$$

$$t - \frac{1}{t} = z$$

$$\left(1 + \frac{1}{t^2}\right) dt = dz$$

$$\Rightarrow \int \frac{dz}{z^2 + (\sqrt{7})^2} = \frac{1}{\sqrt{7}} \tan^{-1}\left(\frac{z}{\sqrt{7}}\right)$$

$$= \frac{1}{\sqrt{7}} \tan^{-1}\left(\frac{\sin x - 1/\sin x}{\sqrt{7}}\right)$$

$$\star \int \frac{f'(x)}{f(x)} \cdot dx = \ln |f(x)| + c$$

$$\star \int [f(x)]^n \cdot f'(x) \cdot dx = \frac{[f(x)]^{n+1}}{n+1} + c$$

Question $\int \frac{dx}{x^4+1}$ Twin Integration

$$\Rightarrow \frac{1}{2} \int \frac{2}{x^4+1} = \frac{1}{2} \int \frac{(x^2+1) - (x^2-1)}{x^4+1} \cdot dx$$

$$\Rightarrow \frac{1}{2} \int \frac{x^2+1}{x^4+1} \cdot dx - \frac{1}{2} \int \frac{x^2-1}{x^4+1} dx$$

$$\Rightarrow \frac{1}{2} \int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} \cdot dx \qquad -\frac{1}{2} \int \frac{1 - 1/x^2}{x^2 + 1/x^2} \cdot dx$$

$x - \frac{1}{x} = t$ $\qquad q = x + \frac{1}{x}$

$$\Rightarrow \frac{1}{2} \int \frac{dt}{t^2 + (\sqrt{2})^2} - \frac{1}{2} \int \frac{dq}{q^2 - (\sqrt{2})^2}$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{\sqrt{2}}\right) \tan^{-1}\left(\frac{t}{\sqrt{2}}\right) - \frac{1}{2\sqrt{2}} \ln\left(\frac{q - \sqrt{2}}{q + \sqrt{2}}\right) + c$$

$$\Rightarrow \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{x - 1/x}{2}\right) - \frac{1}{2\sqrt{2}} \ln\left(\frac{x + \frac{1}{x} - \sqrt{2}}{x + \frac{1}{x} + \sqrt{2}}\right) + c$$

Question $\int \frac{dx}{\cos(x-a)\cos(x-b)}$

$$\Rightarrow \frac{1}{\sin(a-b)} \int \frac{\sin(a-b)}{\cos(x-a)\cos(x-b)} \cdot dx$$

$$\{\sin(a-b) = \sin(x-b) - \sin(x-a)\}$$

$$\Rightarrow \frac{1}{\sin(a-b)} \int \frac{\sin(x-b)\cos(x-a) - \cos(x-b)\sin(x-a)}{\cos(x-a)\cos(x-b)}$$

$$\Rightarrow \frac{1}{\sin(a-b)} \int \left(\tan(x-b) - \tan(x-a)\right) \cdot dx$$

$$\Rightarrow \frac{1}{\sin(a-b)} \left[\ln\left[\sec(x-b)\right] - \ln\left[\sec(x-a)\right]\right] + c$$

Question $\int \frac{dx}{\sin(x-a)\sin(x-b)}$

$$\Rightarrow \frac{1}{\sin(a-b)} \int \frac{\sin\left((x-b)-(x-a)\right)}{\sin(x-a)\sin(x-b)} \cdot dx$$

$$\Rightarrow \operatorname{cosec}(a-b) \left[\left(\cot(x-b) - \cot(x-a)\right) \cdot dx\right.$$

$$\Rightarrow \operatorname{cosec}(a-b) \left[\ln\left|\sin(x-b)\right| - \ln\left|\sin(x-a)\right|\right]$$

STANDARD SUBSTITUTIONS

Sometimes Substitution of value of x convert the whole integrand in some known formate whose integration is comparatively easy.

Following are some standard expression and their substitution :-

- x^2+a^2, $\sqrt{x^2+a^2}$, $\frac{1}{x^2+a^2}$, $\frac{1}{\sqrt{x^2+a^2}}$

  $x = a\tan\theta$
  $x = a\cot\theta$
  $x = a\sec\theta$
  $x = a\operatorname{cosec}\theta$
  $x = a\sin\theta$
  $x = a\cos\theta$
  $x = a\cos 2\theta$

- x^2-a^2, $\sqrt{x^2-a^2}$, etc
- a^2-x^2, $\sqrt{a^2-x^2}$, etc
- $a+x$, $a-x$, $\sqrt{\frac{a+x}{a-x}}$ etc
- $\frac{1}{(x+a)^{1-1/x}(x+b)^{1+\frac{1}{x}}}$ $n \in N$ $n>1$ then substitute $t = \left(\frac{x+a}{x+b}\right)$
- $\left(x \pm \sqrt{x^2 \pm 1}\right)^n$, then substitute $t = \left(x \pm \sqrt{x^2 \pm 1}\right)$

Question $\int \frac{dx}{x^2+a^2}$ $x = a\tan\theta$ $dx = a\sec^2\theta\, d\theta$

$$\Rightarrow \int \frac{a\sec^2\theta\, d\theta}{a^2\sec^2\theta}$$

$$\Rightarrow \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$$

Question $\int \frac{dx}{\sqrt{a^2-x^2}}$ $x = a\sin\theta \Rightarrow dx = a\cos\theta\, d\theta$

$$\Rightarrow \int \frac{a\cos\theta\, d\theta}{a\sqrt{1-\sin^2\theta}} = \int d\theta = \theta + c \Rightarrow \sin^{-1}\left(\frac{x}{a}\right)+c$$

Question

$$\int \frac{dx}{(x-5)^{3/4}\,(x+2)^{5/4}}$$

$$\Rightarrow \int \frac{dx}{(x-5)^{1-\frac{1}{4}}\,(x+2)^{1+\frac{1}{4}}}$$

$$\Rightarrow \int \frac{dx}{(x-5)^{3/4}\,(x+2)^{5/4}\cdot\frac{(x+2)^{3/4}}{(x+2)^{3/4}}}$$

$$\Rightarrow \int \frac{dx}{\left(\frac{x-5}{x+2}\right)^{3/4}(x+2)^2}$$

$$\Rightarrow \int \frac{(x+2)^2\,dt}{7\,t^{3/4}\,(x+2)^2}$$

$$n = \frac{1}{4}$$

$$t = \left(\frac{(x-5)}{x+2}\right)$$

$$\frac{dt}{dx} = \frac{x+2-x+5}{(x+2)^2}$$

$$\frac{dt}{dx} = \frac{7}{(x+2)^2}$$

$$dx = \frac{(x+2)^2}{7}\cdot dt$$

$$\Rightarrow \frac{1}{7}\int t^{-3/4}\,dt = \frac{1}{7}\frac{t^{1/4}}{1/4} = \frac{4}{7}\left[\frac{x-5}{x+2}\right]$$

INTEGRATION BY PART

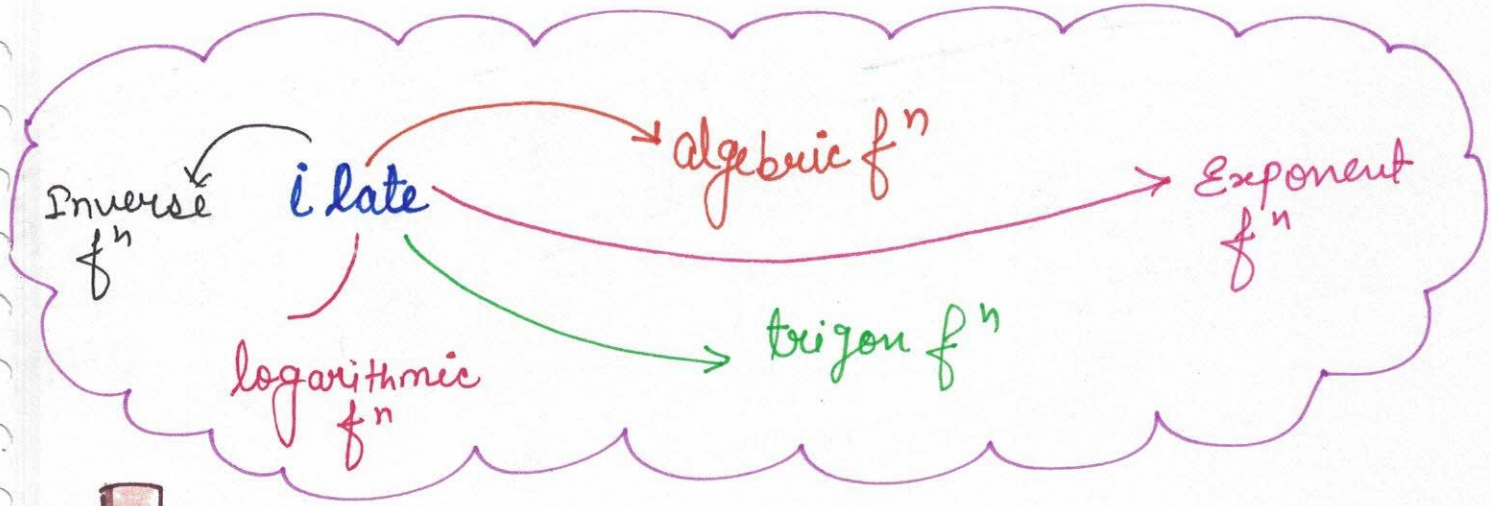
It is a method used to integrate a function which is Product of 2 different natural fun^n

By Part States that -...

$$\int \underset{\text{I}}{f(x)}\cdot\underset{\text{II}}{g(x)}\cdot dx = f(x)\int g(x)\cdot dx - \int\left[\frac{d}{dx}(f(x))\cdot\int g(x)\cdot dx\right]\cdot dx$$

$$= \text{I}\int \text{II}\,dx - \int\left(\frac{d}{dx}(\text{I})\cdot\int(\text{II}\,dx)\cdot dx\right.$$

There, selection of I and II f^n will be done by-



Question

$$\int \underset{I}{x} \underset{II}{\cos x}\, dx = x\int \cos x\, dx - \int 1 \cdot \int \cos x \cdot dx \cdot dx$$

$$= x(\sin x) - \int \sin x\, dx$$

$$= x\sin x + \cos x + c$$

Question

$$\int \underset{I}{x^3} \cdot \underset{II}{e^x} \cdot dx = x^3 \int e^x \cdot dx - \int 3x^2 \cdot e^x\, dx$$

$$= x^3 e^x - 3\left[x^2 e^x + \int 2x e^x\right]$$

$$= x^3 e^x - 3x^2 e^x - 6\left[x e^x - e^x\right] + c$$

$$= x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + c$$

$\int f(x) \cdot g(x) \cdot dx$ $f(x) \rightarrow$ algebric f^n

$g(x) \rightarrow$ integrable any no. of terms

$$\sum_{\lambda=0}^{n-1} (-1)^{\lambda} \left(\lambda^{th} \text{ diff of } f(x)\right)\left[(\lambda+1)^{th} \text{ integration of } g(x)\right.$$

Question $\int x^5 e^{3x}\,dx$

$$= \frac{x^5 e^{3x}}{3} - \frac{5x^4 e^{3x}}{9} + \frac{20x^3 e^{3x}}{27} - \frac{60x^2 e^{3x}}{81}$$

$$\frac{120x e^{3x}}{243} - \frac{120 e^{3x}}{729} + c$$

Question $\int x^2 \sin x\,dx$

$$\Rightarrow x^2(-\cos x) - 2x(-\sin x) + 2(\cos x) + c$$

$$= -x^2\cos x + 2x\sin x + 2\cos x + c$$

Question $\int \underset{II}{e^{ax}} \underset{I}{\sin bx}\cdot dx$

$$= \frac{\sin bx e^{ax}}{a} - \int b\cos bx \cdot \frac{e^{ax}}{a}\cdot dx$$

$$f = \frac{\sin bx e^{ax}}{a} - \left[\frac{b}{a}\right]\left[\frac{\cos bx e^{ax}}{a} - \int \frac{b\sin bx \cdot e^{ax}}{a}\cdot dx\right]$$

$$f = \frac{\sin bx e^{ax}}{a} - \frac{b}{a^2}\cos bx e^{ax} + \frac{b^2}{a^2}\left[\int \sin bx e^{ax}\cdot dx\right.$$

$$f = \frac{\sin bx e^{ax}}{a} - \frac{b}{a^2}\cos bx e^{ax} - \frac{b^2}{a^2}(f)$$

$$f + \frac{b^2}{a^2} I = \frac{\sin bx e^{ax}}{a} - \frac{b}{a^2}\cos bx e^{ax}$$

$$f = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2}$$

USE OF INTEGRATION BY PARTS

USE-I If we have to integrate $f(x)$ w.r.t. x where the differentiation of $f(x)$ is a known quantity, then

$$\int f(x)\cdot dx = \int \underset{\text{I}}{f(x)}\cdot \underset{\text{II}}{1}\cdot dx$$

Ex:- $$\int \ln x\cdot dx = \ln x\cdot \int dx - \int\left(\frac{d}{dx}(\ln x)\cdot \int dx\right)\cdot dx$$

$$= x\ln x - x + c = x\ln\left(\frac{x}{e}\right) + c$$

Question $\int \tan^{-1}x\cdot dx$

$$= x\tan^{-1}x - \int \frac{1}{1+x^2}\cdot x\,dx$$

$1+x^2 = t$

$2x\,dx = dt$

$$= x\tan^{-1}x - \int \frac{dt}{2t}$$

$$= x\tan^{-1}x - \frac{1}{2}\ln t + c$$

$$\Rightarrow x\tan^{-1}x - \frac{1}{2}\ln(1+x^2) + c$$

Question $\int \sqrt{a^2-x^2}\cdot dx$

$$I = x\sqrt{a^2-x^2} - \int \frac{1}{2\sqrt{a^2-x^2}}(-2x)\cdot x\,dx$$

$$I = x\sqrt{a^2-x^2} + \int \frac{x^2+a^2-a^2}{\sqrt{a^2-x^2}}\cdot dx$$

$$I = x\sqrt{a^2-x^2} + \left[-\int \sqrt{a^2-x^2} + \frac{a^2}{\sqrt{a^2-x^2}}\cdot dx\right]$$

$$2I = x\sqrt{a^2-x^2} + a^2\int \frac{dx}{\sqrt{a^2+x^2}}$$

$$I = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$$

Question $\int \sqrt{x^2+a^2}\cdot dx$

$$I = x\sqrt{x^2+a^2} - \int \frac{1}{2\sqrt{x^2+a^2}}(2x)\cdot x\,dx$$

$$I = x\sqrt{x^2+a^2} - \int \frac{x^2+a^2}{\sqrt{x^2+a^2}} - \frac{a^2}{\sqrt{x^2+a^2}}\cdot dx$$

$$I = x\sqrt{x^2+a^2} - \left[\int \sqrt{x^2+a^2}.dx + \int \frac{a^2}{\sqrt{x^2+a^2}}\cdot dx\right.$$

$$2I = x\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left(x+\sqrt{x^2+a^2}\right)$$

USE - 2 $\int e^x\left(f(x)+f'(x)\right)\cdot dx = e^x\left(f(x)\right)+c$

Question $\int e^x\left(\ln x+\frac{1}{x}\right)\cdot dx$

$e^x(\ln x)+c$

Question $\int e^x(\sin x+\cos x)\cdot dx$

$e^x(\sin x)+c$

Question $\int e^x \sin x\,dx = \frac{1}{2}\int e^x(2\sin x)\cdot dx$

$$= \frac{1}{2}\int e^x\left(\underbrace{\sin x+\cos x}_{f'(x)} + \underbrace{\sin x-\cos x}_{f(x)}\right)\cdot dx$$

$$= \frac{1}{2}\cdot e^x(\sin x-\cos x)+c$$

Question $\int e^x \left(\frac{x-1}{x^2+1}\right)^2 \cdot dx$

$$= \int \frac{e^x (x^2+1-2x)}{(x^2+1)} \cdot dx \Rightarrow \int e^x \left[\frac{x^2+1}{(x^2+1)^2} - \frac{2x}{(x^2+1)^2}\right] \cdot dx$$

$$= \int e^x \left(\frac{1}{x^2+1} - \frac{2x}{(x^2+1)^2}\right) \cdot dx \Rightarrow \boxed{e^x \left(\frac{1}{x^2+1}\right) + c}$$

Question $\int e^x \left(\frac{x-1}{x^2}\right) \cdot dx$

$$= \int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) \cdot dx = \boxed{e^x \left(\frac{1}{x}\right) + c}$$

Question $\int e^x \frac{(x-1)}{(x+1)^3} \cdot dx$

$$\int e^x \left[\frac{(x+1)}{(x+1)} - \frac{2}{(x+1)^3}\right] \cdot dx = \int e^x \left[\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}\right] \cdot dx$$

$$\boxed{= \frac{e^x}{(x+1)^2} + c}$$

Question $\int e^x (1 - \cot x + \cot^2 x)\, dx$

$$\int e^x (\operatorname{cosec}^2 x - \cot x) \cdot dx = -\int e^x (\cot x - \operatorname{cosec}^2 x) \cdot dx$$

$$\boxed{= -e^x (\cot x) + c}$$

Question $\int e^{2x} (2 \sin 3x + 3 \cos 3x) \cdot dx$

$2x = t$ $\qquad 2dx = dt \qquad \Rightarrow dx = dt/2$

$$I = \int e^t \left[2\sin 3(t/2) + 3\cos(3t/2)\right] \cdot \frac{dt}{2}$$

$$I = \frac{1}{2}(e^t)\left(2\sin\frac{3t}{2}\right) + c = e^t \sin\left(\frac{3t}{2}\right) + c$$

Question $\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right) \cdot dx$

$\ln x = t \qquad x = e^t \qquad \frac{1}{x}dx = dt \qquad dx = x\,dt = e^t dt$

$$I = \int \left(\frac{1}{t} - \frac{1}{t^2}\right) \cdot e^t dt \Rightarrow e^t\left(\frac{1}{t}\right) \Rightarrow \boxed{\frac{x}{\cos x} + c}$$

Question $\int \left[\frac{(\ln x - 1)}{(\ln x)^2 + 1}\right]^2 \cdot dx$

$\ln x = t \qquad x = e^t \qquad dx = e^t dt$

$$I = \int \left(\frac{(t-1)}{t^2+1}\right)^2 \cdot e^t dt \Rightarrow \int \frac{t^2+1-2t}{(t^2+1)^2} \cdot e^t dt$$

$$I = \int \left[\frac{1}{(t^2+1)} - \frac{2t}{(t^2+1)^2}\right] e^t dt$$

$$= e^t\left(\frac{1}{t^2+1}\right) + c \Rightarrow \boxed{\frac{x}{\ln^2 x + 1} + c}$$

Question $\int \sin(\ln x) \cdot dx$

$\ln x = t \qquad x = e^t \qquad dx = e^t dt$

$$I = \int \sin t \cdot e^t dt = \frac{1}{2}\int e^t (2\sin t)\, dt$$

$$= \frac{1}{2}\int e^t (\sin t + \cos t + \sin t - \cos t)\, dt$$

$$= \frac{1}{2} \cdot e^t (\sin t - \cos t) + c \Rightarrow \boxed{\frac{1}{2} \cdot x(\sin \ln x - \cos \ln x) + c}$$

USE - 3

$$\int \frac{f(x)}{[g(x)]^n}\cdot dx = \int \frac{f(x)}{g'(x)}\cdot\frac{g'(x)}{[g(x)]^n}\cdot dx$$

I fun^n IInd fun^n

Question

$$\int \frac{x^2}{(x\sin x+\cos x)^2}\cdot dx$$

$$\Rightarrow \int \frac{x^2}{x\cos x}\cdot\frac{x\cos x}{(x\sin x+\cos x)^2}\cdot dx$$

$$\Rightarrow \int \underset{I}{x\sec x}\cdot\underset{II}{\frac{x\cos x}{(x\sin x+\cos x)^2}}\cdot dx$$

$$= x\sec x\int\frac{x\cos x}{(x\sin x+\cos x)^2}\cdot dx = -\int\left[(x\sec x\tan x+\sec x)\cdot\int\frac{x\cos x}{(x\sin x+\cos x)^2}\cdot dx\right]$$

$$= x\sec x\left(\frac{-1}{x\sin x+\cos x}\right)-\int\sec x(x\tan x+1)\left(-\frac{1}{x\sin x+\cos x}\right)\cdot dx$$

$$= x\sec x\left(\frac{-1}{x\sin x+\cos x}\right)-\int\sec x\left(\frac{x\sin x+\cos x}{\cos x}\right)\left(\frac{-1}{x\sin x+\cos x}\right)\cdot dx$$

$$= \frac{-x\sec x}{(x\sin x+\cos x)}-\int\sec^2 x\,dx$$

$$= \frac{-x\sec x}{(x\sin x+\cos x)}-\tan x+c$$

PARTIAL FRACTION

TYPE-I If the integrand is in the form of $\int \frac{f(x)}{g(x)} \cdot dx$

$f(x)$ and $g(x)$ both are polynomial.

$g(x)$ is product of some linear expression and degree of $f(x)$ is less than degree of $g(x)$.

Question $\int \frac{3x+5}{(x+1)(x+2)} \cdot dx$

$$\frac{3x+5}{(x+1)(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+2)} = \frac{A(x+2)+B(x+1)}{(x+1)(x+2)}$$

$$3x+5 = x(A+B) + 2A+B$$

$$3 = A+B \qquad 5 = 2A+B$$

$$A = 2 \quad , \quad B = 1$$

$$I = \int \frac{2}{(x+1)} + \frac{1}{(x+2)} \cdot dx$$

$$\boxed{I = 2\ln(x+1) + \ln(x+2) + c}$$

Question $\int \frac{x^2+1}{(x+1)(2x+5)(x+2)} \cdot dx$

$$I = \frac{A}{x+1} + \frac{B}{2x+5} + \frac{C}{x+2}$$

$$A = \frac{2}{3} \qquad B = \frac{29}{3} \qquad C = -5$$

$$\boxed{I = \frac{2}{3}\ln(x+1) + \frac{29}{6}\ln(2x+5) - 5\ln(x+2) + c}$$

Question $\int \frac{2\sin 2x + 3\cos x}{\sin^2 x - 3\sin x + 2} \, dx$

$\sin x = t \qquad \cos x \, dx = dt$

$\int \frac{2\sin x \cos x + 3\cos x}{\sin^2 x - 3\sin x + 2} \, dx$

$I = \int \frac{(4t+3)\,dt}{t^2 - 3t + 2} = \int \frac{4t+3}{(t-2)(t-1)} \, dt$

$\frac{A}{t-2} + \frac{B}{t-1} \qquad A = \frac{11}{1} \Rightarrow B = -7$

$I = 11\ln(t-2) - 7\ln(t-1) + C$

$I = 11\ln(\sin x - 2) - 7\ln(\sin x - 1) + c$

TYPE - II

$\int \frac{f(x)}{g(x)} \, dx \qquad f(x), g(x) \rightarrow$ Polynomial

$g(x)$ is Product of some linear expression

degree of $f(x) \geq$ degree of $g(x)$

In order to obtain the soln of such question first divide $f(x)$ by $g(x)$ in such a way that degree of $f(x) <$ degree of $g(x)$

Question $\int \frac{x\,dx}{(x-1)^2(x-2)}$

$\frac{x}{(x-1)^2(x-2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2}$

$x = A(x^2 - 3x + 2) + B(x-2) + C(x^2 - 2x + 1)$

$A + C = 0 \qquad 3A + B - 2C = 1$

$2A - 2B + C = 0$

$$-3A + B + 2A = 1$$
$$2A - 2B - A = 0$$

$A = -2 \qquad B = -1 \qquad C = 2$

Question $\int \frac{3x+5}{(x+1)^2(x-2)} \cdot dx$

$$= \frac{A}{(x+1)} + \frac{B}{(x+1)^2} + \frac{C}{(x-2)} = \frac{3x+5}{(x+1)^2(x-2)}$$

$$A(x^2-x+2) + B(x-2) + C(x^2+1+2x) = 3x+5$$

$A + C = 0 \qquad A = -C$

$$2A - 2B + C = 5$$
$$-2B - C = 5$$
$$A = \frac{-11}{5}$$

$$-A + B + 2C = 3$$
$$3C + B = 3$$
$$6C + 2B = 6$$
$$3C - 2B = 5$$
$$7C = 11$$
$$C = \frac{11}{7}$$

$$B = \left(5 + \frac{11}{5}\right) = \frac{-36}{10} = \frac{-18}{5} = \frac{-2}{3}$$

$$\int \frac{B}{(x+1)^2} = B \int \frac{x+1}{(x+1)^2} - \frac{x}{(x+1)^2}$$

$$\frac{-11}{9}\ln(x+1) - \frac{2}{3}\int \frac{1}{(x+1)^2} \cdot dx \; \frac{-11}{9}\ln(x-2)$$

TYPE IV $\int \frac{f(x)}{g(x)}$; $f(x), g(x) \longrightarrow$ Polynomial

$g(x)$ is Product of linear exp.

as well as some quadratic exp. which cannot be factorial

degree of $f(x)$ < degree of $g(x)$

Question $\int \frac{3x+2}{(x+1)(x^2+5)} \cdot dx = \int \left(\frac{A}{x+4} + \frac{Bx+C}{x^2+5} \right) dx$

$$3x+2 = Ax^2+5A+Bx^2+Bx+Cx+C$$

$A+B=0 \qquad B+C=3 \qquad 5A+C=2$

$A=-\frac{1}{6} \qquad B=\frac{1}{6} \qquad C=\frac{7}{6}$

$$\Rightarrow \frac{-1}{6}\ln(x+1)+\frac{1}{6}\int \frac{x+17}{x^2+5}\,dx$$

$$\Rightarrow -\frac{1}{6}\ln(x+1)+\frac{1}{6}\left[\frac{1}{2}\int \frac{2x}{x^2+5}dx+\int \frac{17}{x^2+(\sqrt{5})^2}\cdot dx\right]$$

$$\Rightarrow -\frac{1}{6}\ln(x+1)+\frac{1}{12}\ln|x^2+5|+\frac{17}{16\sqrt{5}}\tan^{-1}\left(\frac{x}{\sqrt{5}}\right)+C$$

Question $\int \frac{(x+3)\cdot dx}{(x^2+3)(x-1)}$

$$\frac{x+3}{(x^2+3)(x-1)} = \frac{A}{(x-1)} + \frac{Bx+C}{(x^2+3)}$$

$$x+3 = A(x^2+3)+(Bx+C)(x-1)$$

$$x+3 = Ax^2+3A+Bx^2+Cx-C$$

$A+B=0 \qquad -B+C=1 \qquad 3A-C=3$

$A=-B \qquad A+C=1 \qquad A=1$

$A=1 \qquad B=-1 \qquad C=0$

TYPE -V

$\int \frac{dx}{x^m(ax^n+b)^p}$ Take x^{np} common & then Proceed

Put. $a+bx^{-n}=t$

Question

$$\int \frac{dx}{x^{19}(x^6+1)} = \int \frac{dx}{x^{25}(1+x^{-6})}$$

if $x^{-6}=t \Rightarrow -6x^{-7}\cdot dx = dt$

$\left\{\frac{1}{x^6}=t-1\right\}$

$$I = \frac{-1}{6}\int \frac{dt}{t}(t^3-1)$$

$$I = \frac{-1}{6}\int\left(\frac{t^8-3t^2+3t-1}{t}\right)\cdot dt$$

$$I = \frac{-1}{6}\left[\frac{t^3}{3}-\frac{3t^2}{2}+3t-\ln|t|+c\right]+c$$

Question

$\int \frac{dx}{x^3(x^2+1)^2}$ $\quad \frac{1}{x^2}=t-1$

$\int \frac{dx}{x^7(x^{-2}+1)}$ $\quad 1+\frac{1}{x^2}=t$ $\quad \frac{-2}{x^3}\cdot dx = dt$

$$= \frac{-1}{2}\int \frac{dt\,(t-1)^2}{t^2} = \frac{-1}{2}\int\left(\frac{t^2}{t^2}-\frac{2t}{t^2}+\frac{1}{t^2}\right)\cdot dt$$

$$= \frac{-1}{2}\left[t-2\ln t+\ln t\right]+c$$

$$= \frac{-1}{2}\left[(1+x^{-2})-2\ln(1+x^{-2})+1(1+x^{-2})\right]+c$$

TYPE VI

METHOD OF LONG DIVISION

Question

$\int \frac{3x+5}{x^3(x^2+x+1)}\cdot dx$

$$\frac{3x+5}{x^3(x^2+x+1)} = \frac{A}{x}+\frac{B}{x^2}+\frac{C}{x^3}+\frac{Dx+E}{x^2+x+1}$$

$$\frac{2x+3}{x^2+1} = 3+2x-3x^2+\frac{3x^5-3x^3}{x^2+1}$$

$$\frac{2x+3}{x^4(x^2+1)} = \frac{3}{x^4}+\frac{2}{x^3}-\frac{2}{x^2}+\frac{2}{x}+\frac{2x-3}{x^2+1}$$

$$I = \frac{-3x^{-3}}{3}-\frac{2x^{-2}}{2}-\frac{2x^{-1}}{-1}-2\ln x-\int\frac{2x+3}{x^2+1}\,dx$$

ALGEBRIC INTEGRATION

ROTIONAL INTEGRATION

TYPE-I

$$\int\frac{dx}{ax^2+bx+c}$$

To solve such question make Perfect square and use the form.

$$\int\frac{dx}{x^2+a^2},\quad \int\frac{dx}{x^2-a^2},\quad \int\frac{dx}{a^2-x^2}$$



$$\int\frac{dx}{x^2+x+1} = \int\frac{dx}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$= \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{x+1/2}{\sqrt{3}/2}\right)+c$$

$$= \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+c$$

Question

$$\int \frac{\cos x\,dx}{\cos^2 x + \sin x}$$

$$\int \frac{\cos x\,dx}{-\sin^2 x + \sin x + 1}$$

$\sin x = t$

$\cos x\,dx = dt$

$$= \int \frac{dt}{-t^2 + t + 1} = -\int \frac{dt}{t^2 - t - 1}$$

$$= -\int \frac{dt}{\left(t - \frac{1}{2}\right)^2 - \frac{5}{4}} = \frac{1}{2}\cdot\frac{2}{\sqrt{5}} \ln\left(\frac{t - \frac{1}{2} - \sqrt{5}/2}{t - \frac{1}{2} + \sqrt{5}/2}\right)$$

$$= \frac{-1}{\sqrt{5}} \ln\left(\frac{2\sin x - 1 - \sqrt{5}}{2t - 1 + \sqrt{5}}\right)$$

TYPE-II

$$\int \frac{ax + b}{cx^2 + dx + 2} \cdot dx$$

$$ax + b = A(2cx + d) + B$$

$$I = \int \frac{A(2cx + d)}{cx^2 + dx + e} + \frac{B}{cx^2 + dx + e} \cdot dx$$

$$I = A \ln(cx^2 + dx + e) + B \int \underbrace{\frac{1}{cx^2 + dx + e}}_{\text{Type - I}} \cdot dx$$

Question

$$\int \frac{3x + 5}{x^2 + 4x + 7} \cdot dx$$

$$3x + 5 = A(2x + 4) + B$$

$$A = \frac{3}{2} \qquad 4\left(\frac{3}{2}\right) + B = 5 \qquad B = -1$$

$$\frac{3}{2} \ln(x^2 + 4x + 7) - 1 \int \frac{1}{(x+2)^2 + (\sqrt{3})^2} \cdot dx$$

$$= \frac{3}{2}\ln(x^2+4x+7) - \frac{1}{\sqrt{3}}\ln^{-1}\left(\frac{x+2}{\sqrt{3}}\right)+c$$



$$\int \frac{x}{x^2+1}\cdot dx \Rightarrow \frac{1}{2}\ln|x^2+1| + c$$

$$x = A(2x)+B$$

$$A = \frac{1}{2} \quad B = 0$$



$$\int \frac{ax^2+bx+c}{ex^2+fx+g}\cdot dx$$

$$(ax^2+bx+c) = A(ex^2+f(x)+g)+B(2ex+f)+c$$

$$I = A\int dx + B\int \frac{2ex+f}{ex^2+fx+g}\cdot dx + C\int \frac{dx}{ex^2+f(x)+g}$$



$$\int \frac{x^2+1}{x^2-x+1}\cdot dx$$

$$\left(x-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{5}}{2}\right)^2 - 1 + B = 2A = 1 \qquad B = 0$$

$$x^2+1 = A(x^2-x+1) + B(2x-1) + C$$

$$A - B + C = 1 \Rightarrow C = 1$$

$$A = 1 \qquad B = 1 \qquad C = 1$$

$$I = x + \ln(x^2-x+1) + \frac{2}{\sqrt{5}}\tan^{-1}\left(\frac{x-1/2}{\sqrt{5}/2}\right)$$



$$\int \frac{x^2+2x+3}{x^2-5x+4}\cdot dx$$

$$x^2+2x+3 = A(x^2-5x+4) + B(2x+2) + C$$

$$A = 1 \qquad -5+2B = 2 \qquad B = 7/2$$

$$4x+7+C = 3 \Rightarrow C = -8$$

$$x + 7/2\ln(x^2-5x+4) - 8$$

Question $\int \frac{e^{3x}+e^{2x}+e^{x}}{e^{2x}-e^{x}+1}\,dx$

Question $\int \frac{x^2-x-2}{x^3-6x^2+11x-6}\,dx$

Question $e^x = t \qquad e^x\,dx = dt$

$$\int \frac{t^3+t^2+t}{t^2-t+1}\cdot\left(\frac{dt}{t}\right) = \int \frac{t^2+t+1}{t^2-t+1}\,dt$$

$$t^2+t+1 = A(t^2-t+1)+B(2t-1)+C$$

$A = 1$

$-A+2B = 1$

$2B = 2$

$B = 1$

$A-B+C = 1$

$1-1+C = 1$

$C = 1$

$$I = x + \ln(t^2-t+1) + \int \frac{1}{\left(\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$I = x + \ln|t^2-t+1| + \left(\frac{2}{\sqrt{3}}\right)\tan^{-1}\left(\frac{t-1/2}{\sqrt{3}/2}\right)+C$$

Question $x^3-6x^2+11x+6$

$$I = \int \frac{x^2-x-2}{(x-1)(x-2)(x-3)}\,dx$$

IRROTIONAL INTEGRATION

TYPE-I $\int \frac{dx}{\sqrt{ax^2+bx+c}}$ or $\int \sqrt{ax^2+bx+c}\cdot dx$

Make perfect Square....

Question $\int \frac{dx}{\sqrt{x^2+4x+7}}$

$$= \int \frac{dx}{\sqrt{(x+2)^2+(\sqrt{3})^2}}$$

$$= \ln\left[(x+2)+\sqrt{(x+2)^2+(\sqrt{3})^2}\right] + c$$

Question $\int \frac{dx}{\sqrt{2+2x-x^2}} = \int \frac{dx}{\sqrt{-(x-2)^2+3}}$

$$-(x^2-2x)+2 = \sin^{-1}\left(\frac{x-1}{\sqrt{3}}\right) + c$$

Question $\int \sqrt{x^2-4x+7}\,dx$

$$\Rightarrow \int \sqrt{(x-2)^2+(\sqrt{3})^2}\cdot dx$$

$$= \left(\frac{x-2}{2}\right)\sqrt{(x-2)^2+(\sqrt{3})^2} + \frac{3}{2}\ln\left((x-2)+\sqrt{(x-2)^2+c}\right.$$

TYPE-II $\int \frac{ax+b}{\sqrt{cx^2+dx+e}}\cdot dx$ or $\int (ax+b)\sqrt{cx^2+dx+e}\,dx$

$$ax+b = A(2cx+d)+B$$

$$I = A\int \frac{(2cx+d)}{\sqrt{cx^2+dx+e}}\cdot dx + B\int \underbrace{\frac{1}{\sqrt{cx^2+dx+c}}}_{\text{Type I}}\cdot dx$$

$$I = A\int (2x+d)\sqrt{ax^2+dx+e}\,dx + B\int \sqrt{cx^2+dx+e}\cdot dx$$

Question

$$\int \left(\sqrt{x^2+2x+5}\right)(x+1)\cdot dx$$

$$x+1 = A(2x+2) + B$$

$A = 1/2$ $\qquad 2(\frac{1}{2}) + B = 1$ $\qquad B = 0$

$$I = \frac{1}{2}\int (2x+2)\sqrt{x^2+2x+5}\cdot dx$$

$$I = \frac{1}{2}\int \sqrt{t}\cdot dt = \frac{\frac{1}{2}(t)^{3/2}}{3} = \frac{(x^2+2x+5)^{3/2}}{3+0}$$

Question

$$\int \frac{x-5}{\sqrt{4x^2+2x}}\cdot dx$$

$$x-5 = A(-2x+2) + B$$

$A = -\frac{1}{2}$ $\qquad 2(\frac{-1}{2}) + B = -5$ $\qquad B = -4$

$$I = \frac{-1}{2}\frac{(4-x^2+2x)^{1/2}}{-1/2} - 4\int \sqrt{4-(x-1)^2+1}$$

$$I = -\frac{1}{2}(4 - x^2+2x)^{1/2} - 4\sin^{-1}\left(\frac{x-1}{\sqrt{5}}\right) + c$$

TYPE - II

$$\int \frac{dx}{(ax+b)\sqrt{cx^2+dx+e}}$$

$$ax+b = \frac{1}{t} \qquad x = \left(\frac{1}{t} - b\right)\frac{1}{a}$$

$$a\,dx = -\frac{1}{t^2}\cdot dt.$$

$$I = -\frac{1}{a}\int \frac{dt\,/t^2}{1/t\,\sqrt{c\left(\frac{1}{a}\left(\frac{1}{t}-b\right)\right)^2 + d\left(\frac{1}{t}-b\right)1/a} + c}$$

Question $\int \frac{dx}{(x+1)\sqrt{x^2+x+1}}$

$x+1 = \frac{1}{t}$ $\quad x = \frac{1}{t} - 1$

$dx = -\frac{1}{t^2}\,dt$ $\quad I = -\int \frac{dt/t^2}{(1/t)\sqrt{\left(\frac{1-t}{t}\right)^2 + \frac{(1-t)t+t^2}{t^2}}}$

$$I = -\int \frac{dt}{\sqrt{1+t^2-2t+t-t^2+t^2}}$$

$$I = -\int \frac{dt}{\sqrt{t^2-t+1}}$$

$$I = -\int \frac{dt}{\sqrt{(t-1/2)^2+(\sqrt{3}/2)^2}} = \boxed{-\ln\left|(t-\tfrac{1}{2})+\sqrt{t^2-t+1}\right| + C}$$

Question $\int \frac{dx}{(2x+3)\sqrt{x^2+x+1}}$

$2x+3 = \frac{1}{t}$ $\quad x = \left(\frac{1}{t} - 3\right)\frac{1}{2}$

$2dx = -\frac{1}{t^2}\,dt$ $\quad I = -\frac{1}{2}\int \frac{dt/t^2}{1/t^2\sqrt{\frac{1}{4}(1-3t)^2+2(1-3t)+4}}$

$$I = \frac{-1}{2\times 2}\int \frac{dt}{\sqrt{1+9t^2-6t+2t-6t^2+4t^2}}$$

$$= \frac{-1}{\sqrt{7}}\int \frac{dt}{\sqrt{t^2-\frac{4t}{7}+\frac{1}{7}}}$$

$$= \frac{-1}{\sqrt{7}}\int \frac{dt}{\sqrt{(t-\frac{2}{7})^2+(\sqrt{3}/7)^2}}$$

$$= \boxed{\frac{-1}{\sqrt{7}}\ln\left|(t-\tfrac{2}{7})+\sqrt{(t-\tfrac{2}{7})^2+\left(\tfrac{\sqrt{3}}{7}\right)^2}+C\right|}$$

TYPE-IV

$$\int \frac{(px+q)\,dx}{(ax+b)\sqrt{cx^2+dx+e}}$$

$$px+q = A(ax+b) + B$$

$$A\int \frac{(ax+b)\,dx}{(ax+b)\sqrt{cx^2+dx+e}} + B\int \frac{dx}{(ax+b)\sqrt{cx^2+dx+e}}$$

↓ Type I

↓ Type III

Question

$$\int \frac{(x+1)\,dx}{(2x+3)\sqrt{x^2+x+1}}$$

$$x+1 = A(2x+3) + B$$

$$A = 1/2 \qquad 3/2 + B = 1 - \frac{3}{2} \Rightarrow B = -\frac{1}{2}$$

$$A\int \frac{dx}{\sqrt{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2}} - \frac{1}{2}\left(\frac{-1}{\sqrt{7}}\right) \ln\left|\left(t-\frac{2}{7}\right) + \sqrt{(t-\frac{2}{7})^2 + (\frac{\sqrt{3}}{7})^2}\right|$$

$$\Rightarrow \frac{1}{2}\ln\left[x+\frac{1}{2}\sqrt{(x+\frac{1}{2})^2+(\frac{\sqrt{3}}{7})^2} + \frac{1}{2\sqrt{7}}\ln\right.$$

TYPE-V

$$\int \frac{(ax^2+bx+c)\,dx}{(dx+e)\sqrt{fx^2+gx+h}}$$

$$ax^2+bx+c = A(dx+e)(2fx+g) + B(dx+e) + C$$

↓ Type I

$$I = A\int \frac{(dx+e)(2fx+g)\,dx}{(dx+e)\sqrt{fx^2+gx+h}} + B\int \frac{dx+e}{(dx+e)\sqrt{fx^2+gx+h}}\,dx$$

Type V

$$+C\int \frac{dx}{(dx+e)\sqrt{fx^2+gx+h}}$$

Type III

TYPE-VI

$$\int \frac{x\,dx}{(ax^2+b)\sqrt{cx^2+d}}$$

In order to obtain the integral of above integrand

Substitute $cx^2+d = t^2$

$$cx^2+d = t^2$$

$$cx\,dx = t\,dt$$

$$x\,dx = \frac{t\,dt}{c}$$

$$x^2 = \frac{t^2-d}{c}$$

$$I = \frac{1}{c}\int \frac{t\,dt}{\left(a\left(\frac{t^2-d}{c}\right)+b\right)t}$$

Can Solve after this



$$\int \frac{x\,dx}{3x^2+5\left(\sqrt{x^2+5}\right)}$$

x^2+t^2-5

$$x^2+5 = t^2$$

$$x\,dx = t\,dt$$

$$I = \int \frac{t\,dt}{(3(t^2-5)+5)\cdot t}$$

$$I = \frac{1}{3}\int \frac{dt}{(t^2-5)+5/3}$$

$$I = \frac{1}{3}\int \frac{dt}{t^2-\left(\sqrt{\frac{10}{3}}\right)^3}$$

$$\Rightarrow I = \frac{1}{3}\times\frac{1}{2}\frac{\sqrt{3}}{\sqrt{10}}\ln\left(\frac{t-\sqrt{10/3}}{t+\sqrt{10/3}}\right)$$

Can Solve after this

TYPE-VII

$$\int \frac{dx}{(ax^2+b)\sqrt{cx^2+d}}$$

$$x = \frac{1}{t} \quad \Rightarrow \quad dx = -\frac{1}{t^2}\,dt$$

$$\int \frac{-dt/t^2}{\left(\frac{a+bt^2}{t^2}\right)\sqrt{\frac{c+dt^2}{t^2}}} = -\int \frac{t\,dt}{(bt^2+a)\sqrt{dt^2+c}}$$

$dt^2+c = z^2$
Known ~~after~~ Method after this

Question $\int \frac{dx}{(3x^2+5)\sqrt{x^2+5}}$

$$x = \frac{1}{t} \qquad dx = -\frac{1}{t^2}\,dt$$

$$-\int \frac{dt/t^2}{\left(\frac{3+5t^2}{t^2}\right)\sqrt{\frac{1+5t^2}{t}}} = -\int \frac{t\,dt}{(3+5t^2)\sqrt{1+5t}}$$

$$1+5t^2 = z^2$$

$$5t^2 = z^2-1$$

$$5t\,dt = 2z\,dz$$

$$\sqrt{\frac{5}{x^2}+1} = z$$

$$t\,dt = \frac{z\,dz}{5}$$

$$I = -\frac{1}{5}\int \frac{z\,dz}{(3+z^2-1)z} = -\frac{1}{5}\int \frac{dz}{z^2+(\sqrt{2})^2}$$

$$= -\frac{1}{5}\cdot\frac{1}{2\sqrt{2}}\tan^{-1}\left(\frac{z}{\sqrt{2}}\right)+c$$

TRIGNOMETRIC INTEGRATION

TYPE-I

CONVERT PRODUCT INTO SUM

Question $\frac{1}{2}\int 2\cos x \cdot \cos 2x \cdot dx$

$$= \frac{1}{2}\int \cos 3x + \cos x \cdot dx \Rightarrow \frac{1}{2}\left[\frac{\sin 3x}{3} + \sin x\right]+c$$

Question

$$\int \cos x \cdot \cos 2x \cdot \cos 4x \cdot dx$$

$$= \frac{1}{2}\int \left[\cos 3x + \cos x\right] \cdot \cos 4x \cdot dx$$

$$= \frac{1}{2}\int \frac{1}{2} 2\cos 3x \cos 4x + \frac{1}{2} 2\cos x \cos 4x \cdot dx$$

$$= \frac{1}{2}\cdot\frac{1}{2}\int \left[\cos 7x + \cos x + \cos 5x + \cos 3x\right] \cdot dx$$

$$= \frac{1}{4}\left[\frac{\sin 7x}{7} + \sin x + \frac{\sin 5x}{5} + \frac{\sin 3x}{3}\right] + c$$

TYPE - II

CONVERT POWER INTO MUITIPLE

Question

$$\int \sin^4\left(\frac{x}{3}\right) \cdot dx$$

$$\frac{1}{4}\int \left[1 - \cos\left(\frac{2x}{3}\right)\right]^2 \cdot dx$$

$$= \frac{1}{4}\int \left[1 + \cos^2\frac{2x}{3} + 2\cos\frac{2x}{3} \cdot dx\right]$$

$$= \frac{1}{4}\int 1 - 2\cos\frac{2x}{3} + \frac{1+\cos\frac{4x}{3}}{2} \cdot dx$$

$$= \frac{1}{4}\left[x - \frac{2\sin\frac{2x}{3}}{2/3} + \frac{1}{2}x + \frac{1}{2}\frac{\sin\frac{4x}{3}}{2\cdot 4/3}\right] + c$$

Question

$$\int \cos^4(3x) \cdot dx$$

$$\frac{1}{4}\int (1+\cos 6x)^2 \cdot dx = \frac{1}{4}\int (1 + \cos^2 6x + 2\cos 6x) \cdot dx$$

$$= \frac{1}{4}\left(x + \frac{2\sin 6x}{6}\right) + \frac{1}{4}\int \frac{1+\cos 2x}{2} \cdot dx$$

$$= \frac{1}{4}\left(x + \frac{\sin 6x}{3}\right) + \frac{1}{8}\left[x + \frac{\sin 12x}{12}\right] + c$$

$$= \frac{3x}{8} + \frac{\sin 6x}{12} + \frac{\sin 12x}{96} + c$$

Question

$$\int \sin^6 x\, dx = \int \sin^2 x \cdot \sin^4 x \cdot dx$$

$$= \frac{1}{8}\int (1-\cos 2x)^2 \cdot (1-\cos 2x)$$

$$= \frac{1}{8}(1-\cos 2x)^3 \cdot dx$$

$$= \frac{1}{8}\left(1 - \cos^3 2x - 3\cos 2x(1-\cos 2x)\right) \cdot dx$$

$$= \frac{1}{8}\left[1 - \frac{\cos 6x + 3\cos 2x}{4} - 3\cos 2x + 3\cos^2 2x \cdot dx\right.$$

$$= \frac{1}{8}\left(x - \frac{1}{4}\left(\frac{\sin 6x}{6} + 3\frac{\sin 2x}{2}\right) - 3\frac{\sin 2x}{2} + \left|\frac{3}{2}\left[x + \frac{\sin 4x}{4}\right] + c\right.\right.$$

TYPE - III

If the integrand is in the form of $+c \cdot f^n$ of $\sin x$ and $\cos x$ in such a way that replacing $\sin x$ by $(-\sin x)$ and $\cos x$ by $(-\cos x)$, integrand will not change, then put $\tan x = t$ or $\cot x = t$.

Question

$$\int \frac{\sin^p x}{\cos^{p+2} x\, dx} \cdot dx$$

$\tan x = t \qquad \sec^2 x\, dx = dt$

$$\int \tan^p x \cdot \sec^2 x\, dx = \int t^p \cdot dt$$

$$= \frac{t^{p+1}}{p+1} + c \Rightarrow \frac{(\tan x)^{p+1}}{p+1} + c$$

Question

$$\int \frac{dx}{\sin x \cdot \cos^5 x}$$

$= \tan x = t \qquad \sec^2 x\,dx = dt$

$$= \int \frac{\cos x \cdot dx}{\sin x \cos^6 x} = \int \frac{\sec^2 x\,dx}{\tan x \cdot \cos^4 x}$$

$= \cot x = t \qquad -\operatorname{cosec}^2 x\,dx = dt$

$$= \int \frac{\sin^5 x}{\sin^6 x \cos^5 x} \cdot dx = \operatorname{cosec}^4 x$$

$$\frac{2}{2(\sin x \cos x)\cos^4 x} = \int \frac{(\sec^4 x) \cdot \sec^2 x\,dx}{\tan x}$$

$$= \int \frac{(1+t^2)^2}{t} dt - \int \frac{1+t^2+2t^2}{t} dt$$

$$= \ln t + \frac{t^4}{4} + \frac{2t^3}{3} + c$$

Question

$$\int \frac{dx}{\sqrt{1-\sin^4 x}} = \int \frac{dx}{\cos x \sqrt{1+\sin^2 x}}$$

$\tan x = t \qquad = \int \frac{\sec x\,dx}{\sqrt{1+\sin^2 x}}$

$$= \int \frac{\sec^2 x\,dx}{\sqrt{\sec^2 x + \tan^2 x}} \Rightarrow \int \frac{dt}{\sqrt{1+2t^2}}$$

$$\Rightarrow \frac{1}{2} \cdot \frac{dt}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + t^2}} \Rightarrow \frac{1}{\sqrt{2}} \cdot 1$$

Question

$$\int \left(\frac{\sin^5 x}{\cos^{11} x}\right)^{1/3} \cdot dx \qquad \tan x = t$$

$$\int \left(\frac{\tan^5 x}{\cos^6 x}\right)^{1/3} \cdot dx = \int (\tan^5 x)^{1/3} \cdot \sec^2 x\,dx$$

$$= \int t^{5/3}\,dt$$

$$= \frac{3t^{8/3}}{8} = \frac{3(\tan x)^{8/3}}{8} + c$$

TYPE-IV If integrand is a f^n of $\sin x$ & $\cos x$ and replacing $\sin x$ by $(-\sin x)$, minus comes out from integrand then Substitute $\cos x = t$.



$\cos x = t$
$-\sin x\,dx = dt$

$$\int \sin^5 x\,dx = \int \sin^4 x \cdot \sin x\,dx$$

$$I = -1\int (1-\cos^2 x)^2 \cdot dt$$

$$I = -\int (1-t^2)^2 \cdot dt$$

$$I = -\int 1+t^4-2t^2 \cdot dt$$

$$I = -\left[t + \frac{t^5}{5} - \frac{2t^3}{3}\right] + c$$

$$\boxed{I = -\left[\cos x + \frac{\cos^5 x}{5} - \frac{2\cos^3 x}{3}\right] + c}$$



$$\int \sin^3 x \cos^2 x\,dx = \int \sin^2 x \cos^2 x \cdot \sin x\,dx$$

$\cos x = t$
$-\sin x\,dx = dt$

$$= -\int (1-\cos^2 x)\cos^2 x\,dt$$

$$= -\int t^2 - t^4\,dt$$

$$= -\left[\frac{t^3}{3} - \frac{t^5}{5}\right] + c$$

$$\boxed{= -\left[\frac{\cos 3x}{3} - \frac{\cos^5 x}{5}\right] + c}$$

TYPE-V If integrand is a fn of $\sin x$ and $\cos x$ and replacing $\cos x$ by $(-\cos x)$, minus sign comes out Then Substitute $\sin x = t$.



$$\int \frac{\cos^3 x\,dx}{3+4\sin^2 x} = \int \frac{(1-\sin^2 x)\cdot dt}{3+4t^2} \qquad \cos x\,dx = dt$$

$$= \int \frac{1-t^2}{3+4t^2} \cdot dt \Rightarrow 4\int \frac{dt}{t^2+(\sqrt{3}/2)^2} - \frac{1}{4}\int \frac{4t^2+3-3}{3+4t^2}\,dt$$

$$= \frac{1 \cdot 2}{4\sqrt{3}} \tan^{-1}\left(\frac{2t}{\sqrt{3}}\right) - \int 1 - \frac{3}{3+4t^2} \cdot dt$$

$$= \frac{1}{2\sqrt{3}} \tan^{-1}\left(\frac{2t}{\sqrt{3}}\right) - \left[t - \frac{3 \cdot 2}{4\sqrt{3}} \tan^{-1}\left(\frac{2t}{\sqrt{3}}\right)\right]$$

Question

$$\int \frac{\sin^{3/2}x + \cos^{3/2}x}{\sqrt{\sin^3 x \cos^3 x \sin(x+\lambda)}} \cdot dx$$

$$= \int \frac{\sin^{3/2}x + \cos^{3/2}x}{\sin x \cos x \sqrt{\sin x \cos x \sin(x+\lambda)}} \cdot dx$$

$$= \int \frac{\sqrt{\sin x}}{\cos x \sqrt{\sin x \cos x \sin(x+\lambda)}} + \frac{\sqrt{\cos x}}{\sin x \sqrt{\sin x \cos x \sin(x+\lambda)}}$$

$$= \int \frac{1}{\cos^{3/2} \sin^{1/2}(x+\lambda)}\,dx + \int \frac{1}{\sin^{3/2} \cdot \sin^{1/2}(x+\lambda)}$$

TYPE VI if the integrand is in the form of -

$$\int (\sin x - \cos x) \cdot f(\sin 2x) \cdot dx$$

Put $\sin 2x = (\sin x + \cos x)^2 - 1$

Let $\sin x + \cos x = t$

$t^2 = 1$

TYPE - VII

$$\int (\sin x + \cos x) f(\sin 2x) \cdot dx$$

Put $\sin 2x = 1 - (\sin x - \cos x)^2 =$

Let $\sin x - \cos x = t$

$1 - t^2$

Question $\int \frac{\sin x}{1+3\sin 2x}\,dx = \frac{1}{2}\int \frac{2\sin x}{1+3\sin 2x}\,dx$

$$= \frac{1}{2}\int \frac{\sin x+\cos x+\sin x-\cos x}{1+3\sin 2x}\,dx$$

TYPE VIII

If the integrand is in the form of $\int(a\sin x+b\cos x+c)\,dx$ Then to obtain the sol^n, convert $\sin x$ and $\cos x$ in the half angle of $\tan x$.

$$\sin x = \frac{2\tan x/2}{1+\tan^2 x/2} \qquad \cos x = \frac{1-\tan^2 x/2}{1+\tan^2 x/2}$$

Question $\int \frac{dx}{3\sin x+4\cos x+5}$ $\quad \tan x/2 = t$

$$\int \frac{dt}{3(2t)+4(1-t^2)+5} = \int \frac{dt}{-4t^2+6t+9}$$

$$= -\frac{1}{4}\int \frac{dt}{t^2-\frac{23t}{7}-\frac{9}{4}} = -\frac{1}{4}\int \frac{dt}{(t-\frac{3}{4})^2-\frac{9}{4}-\frac{9}{16}}$$

$$= -\frac{1}{4}\int \frac{dt}{(t-\frac{3}{4})^2-\frac{45}{16}} = -\frac{1}{4}\int \frac{dt}{(t-\frac{3}{4})^2-\frac{45}{16}}$$

$$-\frac{1}{4}\,\frac{1}{2}\sqrt{\frac{16}{45}}\,\ln\left|\frac{(t-3/4)-\frac{\sqrt{45}}{4}}{(t-3/4)+\sqrt{45}/4}\right|$$

Question $\int \frac{dx}{\sin x+\cos x}$ $\quad \tan\frac{x}{2} = t$

$2\sec^2\frac{x}{2}\,dx = dt$

$$\int \frac{dt}{2(1+t^2)\left(\frac{2t+1-t^2}{1+t^2}\right)} = -\frac{1}{2}\int \frac{dt}{t^2-2t-1}$$

$$= -\frac{1}{2}\int \frac{dt}{(t-1)^2-\sqrt{2}^2} = -\frac{1}{2}\times\frac{1}{2\sqrt{2}}\ln\left|\frac{t-1-\sqrt{2}}{t-1+\sqrt{2}}\right|+c$$

TYPE

If the integrand is in the form of $\int \frac{dx}{a\sin^2 x + b\cos^2 x + c\sin x \cos x}$, then to obtain the integral convert the exp. into $\tan x$.

Question

$$\int \frac{\cos x\, dx}{\sin x + \cos x}$$

$$\frac{1}{2}\int \frac{2\cos x\, dx}{\sin x + \cos x} = \frac{1}{2}\int \left(\frac{\cos x + \sin x + \cos x - \sin x}{\sin x + \cos x}\right) dx$$

$$= \frac{1}{2}\int dx + \int \frac{\cos x - \sin x}{\sin x + \cos x}\cdot dx = \frac{x}{2} + \ln(\sin x + \cos x) + c$$

Question

$$\int \frac{dx}{3\sin^2 x + 4\cos^2 x + \sin x \cos x}$$

$\tan x = t$

$$= \int \frac{\sec^2 x\, dx}{3\tan^2 x + \frac{4}{3} + \tan x} = \frac{1}{3}\int \frac{\sec^2 x\, dx}{(\tan x + \frac{1}{6}) + (\frac{\sqrt{17}}{6})^2}$$

$$= \frac{1}{3}\int \frac{dt}{(t+\frac{1}{6})^2 + (\frac{\sqrt{17}}{6})^2} = \frac{2}{\sqrt{17}} \times \frac{1}{3}\tan^{-1}\left(\frac{(t+\frac{1}{6})6}{\sqrt{47}}\right)$$

$$= \frac{6}{3\sqrt{47}}\tan^{-1}\left[\frac{6(\tan x + 1/6)}{\sqrt{47}}\right] + c$$

TYPE

If the integrand is in the form of $\int \sin^m x \cdot \cos^n x \cdot dx$ and $\frac{m+n-2}{2}$ is -ve integer. Put $\tan x = t$

Question

$$\int \sin^{-7/5} x \cos^{-3/5} x\, dx$$

$\tan x = t$

$$\int \tan^{-7/5} x \cos^{-3/5} x \cos^{-7/5} x\, dx \Rightarrow \tan^{-7/5} x \cos^{-2} x\, dx$$

$$= \int t^{-7/5}\, dt = \frac{5t^{-7/5}}{-2} + c$$

Question $\int \sin^{-4/3} x \cdot \cos^{-2/3} x \, dx$

$$\int \tan^{-4/3} x \cdot \cos^{-2/3-4/3} x \, dx$$

$$= \int \tan^{-4/3} x \cos^{-2} x \, dx = \frac{3 \tan^{-1/3} x}{-1}$$

$$= \frac{-3}{(\tan x)^{1/3}} + c$$

Question $\int (\sqrt{\tan x} + \sqrt{\cot x}) \, dx$

$$= \int \frac{\tan x + 1}{\sqrt{\tan x}} \cdot dx$$

$\tan x = 1$

$\sec^2 x \, dx = 2t \, dt$

$(1+t^4) \, dx = 2t \, dt$

$t = \sqrt{\tan x}$

$$= \int \frac{(t^2+1)}{t} \left(\frac{2t \, dt}{1+t^4} \right)$$

$$= 2 \int \frac{t^2+1}{t^4+1} \cdot dt = 2 \int \frac{1 + 1/t^2}{t^2 + \frac{1}{t^2}} \, dt$$

$$= 2 \int \frac{1 + 1/t^2 \cdot dt}{\left(t - \frac{1}{t}\right)^2 + 2}$$

$t - \frac{1}{t} = z$

$1 + \frac{1}{t^2} \, dt = dz$

$$= 2 \int \frac{dz}{z^2 + 2} = 2 \frac{1}{\sqrt{2}} \tan^{-1} \frac{z}{\sqrt{2}}$$

$$= \sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \frac{1}{\sqrt{\tan x}}}{2} \right) + c$$

MISCELLANEOUS

* If the integrand is in the form of $\int x^m (a+bx^n)^p \cdot dx$

$p \neq 0$, Then.

- if p is an natural number, then expand $(a+bx^n)^p$ using binomial Theorem.
- if p is a negative integer and m and n are rational number then substitute $x = t^k$, where k is LCM of denominator of m and n.
- If $\frac{m+1}{n}$ is an integer and p is rational the put $(a+bx^n) = t^k$, k = denominator of p.
- If $\frac{m+1}{n}$ is not an integer and p is a rational number, then substitute $\frac{a+bx^n}{x^n} = t^k$ where k is denominator of p.

Question $\int x^{3/2}(1+x^2)^3 dx = \int x^{3/2}(1+x^6+3x^4+3x^2)\,dx$

$$= \frac{2x^{5/2}}{5} + \frac{2x^{17/2}}{17} + \frac{6x^{13/2}}{13} + \frac{6x^{9/2}}{9} + c$$

Question $\int x^{-2/3}(1+x^{-2/3})^{-1} \cdot dx \qquad x = t^3 \qquad dx = 3t^2 dt$

$$\int (t^{-2})(1+t^{-2})^{-1}(3t^2)\,dt = 3\int (1+t^{-2})^{-1} dt$$

$$= 3\int \frac{t^2+1^{-1}\,dt}{t^2+1} = 3t - \frac{3}{t} + c \qquad = 3t - 3\tan^{-1} t$$

Question $\int x^{-1/3}(1+x^{1/3})^{1/4} \cdot dx \Rightarrow \frac{\frac{-1}{3}+1}{1/3} = 2$ integer

$(1+x^{1/3}) = t \qquad x^{1/3} = t^4 - 1 \Rightarrow \frac{1}{x^{1/3}} = \frac{1}{t^4-1}$

$\frac{1}{3}x^{-2/3}dx = 4t^3dt \Rightarrow \int (t)^4 . 4t^3dt \times 3 \quad (t^4-1)$

$= 12\int t^4(t^4-1)\,dt \Rightarrow \frac{12t^5}{9_2} - \frac{t^5}{5} + c$

$$= \frac{4}{3}\left(1+x^{1/3}\right)^9 - \frac{\left(1+x^{1/3}\right)^5}{5} + c$$



$\int x^{-11}\left(1+x^4\right)^{-1/2}\cdot dx \quad = \frac{1+x^4}{x^4} = t^2 \qquad \frac{1}{x^4} = t^2$

$\frac{1}{x^2} = t$

$\frac{-4}{x^5}dx = 2t\,dt$

$= \int x^{-11}\left(t^{-1}\right)x^4 . \frac{2t\,dt}{2_4} \cdot x^5$

$= \frac{1}{2}\int \frac{t}{x^2 t}\,dt = \frac{1}{2}\int t\,dt \Rightarrow \frac{1}{2}\frac{t^2}{2} = \frac{1}{4}\left(\frac{1+x^4}{x^4}\right) + c$

REDUCTION FORMULA

$I_n = \int \tan^n x\,dx$

$I_n = \int \tan^{n-2} x \tan^2 dx$

$I_n = \int \tan^{n-2} x \left(\sec^2 x - 1\right).dx$

$I_n = \int t^{n-2}dt - \int \tan^{n-2} x\,dx$

$I_n = \frac{t^{n-1}}{n-1} - I_{n-2} + c$

$I_n + I_{n-2} = \frac{t^{n-1}}{n-1} + c$